

Quick Check

Lesson 7: Plant Hormones and Apical Dominance: What Coordinates Growth in Maize and Beans?

8 multiple-choice questions • choose A, B, C or D

Sub-Strand 2.2: Growth and Development in Plants | Lesson 7

Mr. Wafula's Rosecoco beans with the shoot tip intact had 0 branches at Day 25. Beans with the tip removed at Day 20 had 3 lateral branches by Day 32. What is the best prediction for why removing the tip made the beans branch?

- A** The plant got more sunlight on the side buds after the tip was cut
- B** Cutting the tip made the plant absorb more water from the soil
- C** Something made in the shoot tip was stopping the side buds from growing
- D** Side buds only grow when the plant is damaged and needs to heal

Which plant hormone is made in the shoot tip, promotes stem elongation and inhibits the growth of lateral buds?

A Ethylene

B Auxin (IAA)

C Absciscic acid

D Cytokinin

A learner is completing a hormone table. Which hormone is the odd one out because it works as a gas and promotes fruit ripening?

A Gibberellins

B Cytokinin

C Ethylene

D Auxin (IAA)

In Mr. Wafula's experiment, a bean plant with its shoot tip intact had 1 main stem and 0 branches at Day 25. A bean plant with the tip removed at Day 20 had 3 lateral branches by Day 32. Which Claim-Evidence-Reasoning statement is the strongest explanation?

- A** The plant with the tip removed was taller, so it grew more branches
- B** Branches grow after Day 30 in all beans, so the tip made no difference
- C** Cytokinin is made in the shoot tip and stops branching, so there were 0 branches
- D** Auxin from the shoot tip inhibits lateral buds; with the tip removed, 3 branches grew instead of 0, so the tip was the cause

A farmer had 8 healthy-looking bean seeds with hard seed coats that did not germinate. After scarification, 7 of the 8 germinated by Day 7. Which hormone idea best explains why the seeds were dormant before scarification and how they woke up?

- A** Hormones played no part, because only the hard seed coat matters
- B** Absciscic acid maintains dormancy, while gibberellins promote germination
- C** Ethylene kept the seeds dormant and auxin made them germinate
- D** Cytokinin maintains dormancy, while absciscic acid promotes germination

After the lesson, a learner says: 'Each plant hormone does only one job.' Which fact from this lesson shows that this idea is wrong?

- A** Ethylene promotes fruit ripening
- B** Gibberellins promote both seed germination and stem elongation
- C** Cytokinins promote cell division
- D** Absciscic acid maintains dormancy

Learners are revising their Day 1 model of maize and beans. Which labelled arrow should they add to show apical dominance?

- A** Absciscic acid from the lateral buds, promoting stem elongation
- B** Gibberellins from the shoot tip, maintaining dormancy
- C** Cytokinin from the shoot tip, inhibiting lateral buds
- D** Auxin from the shoot tip, inhibiting lateral buds

A farmer in Kitale cuts off the shoot tip of a bean plant and wants more side branches. Which hormone effect explains why this works?

- A** Cutting the tip makes ethylene ripen the side buds into branches
- B** Cutting the tip removes cytokinin, which allows the side buds to grow
- C** Without the tip, less auxin reaches the side buds, so they are no longer inhibited and can grow
- D** Cutting the tip makes the plant produce more abscisic acid, which grows side buds

All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).